

BMP3 Sensor Driver

API Reference

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Overview

The **BMP3** driver provides a portable C API for the Bosch BMP390 and BMP388 barometric pressure and temperature sensors. It handles sensor initialisation, raw data retrieval over I2C or SPI, and compensation of raw ADC values into physical units (Pa and °C) using the factory-calibrated trimming parameters stored in the device NVM.

Key features:

- **Bus-agnostic** – Works over I2C and SPI via user-supplied read/write callbacks.
- **Self-contained** – Single header, no dynamic allocation, C99-compatible.
- **Compensation** – Full Bosch double-precision compensation formula included.
- **Error reporting** – Signed 8-bit return codes for easy chained error checking.

Return Codes

All API functions return `int8_t`. A return value of **BMP3_OK (0)** indicates success; negative values indicate errors.

Macro	Value	Description
BMP3_OK	0	Operation completed successfully.
BMP3_E_NULL_PTR	-1	A required pointer argument was NULL.
BMP3_E_COMM_FAIL	-2	Bus communication (I2C/SPI) failed.
BMP3_E_INVALID_LEN	-3	An invalid data length was supplied.
BMP3_E_DEV_NOT_FOUND	-4	No BMP3 device detected on the bus.

Data Structures

`struct bmp3_calib_data`

Calibration data structure for the BMP3 sensor.

Holds the trimming parameters read from the sensor's NVM. These are used internally for compensating raw ADC readings.

Fields:

Type / Name	Description
double par_t1	Calibration coefficient T1
double par_t2	Calibration coefficient T2
double par_t3	Calibration coefficient T3
double par_p1	Calibration coefficient P1
double par_p2	Calibration coefficient P2
double t_lin	Compensated temperature (linear intermediate)

struct bmp3_uncomp_data

Uncompensated (raw) ADC sensor data.

Populated by bmp3_get_uncomp_data() and passed to bmp3_compensate_data().

Fields:

Type / Name	Description
uint32_t pressure	Raw pressure ADC value
uint32_t temperature	Raw temperature ADC value

struct bmp3_data

Compensated sensor data in SI units.

Populated by bmp3_compensate_data() for application use.

Fields:

Type / Name	Description
double pressure	Compensated pressure in Pascals (Pa)
double temperature	Compensated temperature in degrees Celsius (°C)

Function Reference

Initialise the BMP3 sensor.

```
int8_t bmp3_init(void *dev)
```

Resets the device, reads the chip-ID to confirm the sensor is present, then loads calibration coefficients from NVM into the device structure.

Parameters:

Parameter	Description
[in,out] dev	Pointer to the initialised device structure.

Returns:

BMP3_OK on success; BMP3_E_NULL_PTR, BMP3_E_COMM_FAIL, or BMP3_E_DEV_NOT_FOUND on failure.

Read raw pressure and temperature from the sensor.

```
int8_t bmp3_get_uncomp_data(struct bmp3_uncomp_data *uncomp_data, const void *dev)
```

Issues a forced measurement and retrieves the 24-bit ADC results over the configured bus (I2C or SPI).

Parameters:

Parameter	Description
[out] uncomp_data	Pointer to store the raw ADC values.
[in] dev	Pointer to the device structure.

Returns:

BMP3_OK on success; negative error code on failure.

Compensate raw sensor data.

```
int8_t bmp3_compensate_data(struct bmp3_data *comp_data, const struct bmp3_uncomp_data *uncomp_data)
```

Applies the Bosch double-precision compensation formula to convert raw ADC values to physical units (Pa and °C).

Parameters:

Parameter	Description
[out] comp_data	Pointer to store compensated output (Pa / °C).
[in] uncomp_data	Raw ADC values from bmp3_get_uncomp_data().
[in] calib_data	Calibration coefficients loaded during bmp3_init().

Returns:

BMP3_OK on success; BMP3_E_NULL_PTR if any pointer is NULL.

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